

Abhay Singh

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SUMMARY

Software & AI Engineer with experience building secure, scalable backend systems, AI applications, and automation used by thousands of users. Currently serving as **Secretary** and former **Development Head** of the college's official tech innovation club (CIDC), leading engineering teams and maintaining critical digital infrastructure. Skilled in designing production-grade REST APIs, secure authentication systems, database optimisation, and developing LLM-powered applications, agentic AI systems, and pipelines. Passionate about building reliable, production-ready software that combines modern AI with scalable backend engineering.

EDUCATION

Army Institute of Technology

B.E. in Information Technology — CGPA: **8.77**

Pune, Maharashtra

July 2024 – Present

St. Mary's Inter College

Senior Secondary (CBSE): **90.6%** | Matriculation (CBSE): **83%**

Lucknow, Uttar Pradesh

EXPERIENCE

Army Institute of Technology — Official Website

[Blog](#)

MongoDB · Express · React · Node.js (MERN)

- Designed and developed AIT's official website serving **2,000+ daily users**, implementing a secure authentication system with **Sign-in, Sign-up, Password Recovery, JWT, MFA/TOTP, and CAPTCHA** using **Node.js** and **Express**.
- Automated Password Recovery System:** Developed a reliable password recovery pipeline using **Nodemailer (SMTP)** to generate and deliver cryptographic password-reset tokens, ensuring secure and seamless self-service account recovery.
- Full-Stack Troubleshooting & Admin Dashboard Optimisation:** Diagnosed and resolved critical frontend and backend bugs across the admin dashboard, enhancing administrative panel stability, data rendering efficiency, and administrative workflows.

SendBox — WhatsApp Automation System

[Blog](#)

Cron Jobs · WaSenderAPI · Supabase · Next.js · React-Native

- Engineered a **WhatsApp automation** platform deployed to **1,700+ students** and faculty, enabling secure, scheduled messaging with multi-user support, cron-based automation, and intelligent rate limiting for reliable delivery.
- Built a secure Supabase backend with **Google OAuth, Row-Level Security (RLS)**, and an optimised **CSV/XLSX bulk import pipeline** featuring schema validation and efficient batch processing.

PROJECTS

Customer Support Agent — Agentic RL Environment

[Blog](#)

FastAPI · OpenEnv Framework · OpenAI · Docker · Hugging Face

- Engineered a Gymnasium-compliant **Reinforcement Learning environment** simulating interactive customer support ticket resolution, supporting standard **step, reset, and state APIs** for agent evaluation.
- Designed dynamic policy checks, keyword-based response grading, and custom dense reward shaping to rigorously benchmark LLM and scripted baseline agent performance.

Restaurant Agent — Intelligent Order Processing System

[Blog](#)

FastAPI · LangChain · LangGraph · OpenAI

- Architected an LLM-powered restaurant assistant, compiling a **cyclic workflow graph** that enables the model to autonomously determine user intent and route execution to custom database tools.
- Engineered a serverless **database** integration using Google Sheets API (via **gsread** and **OAuth 2.0** service accounts) to enable the LLM to run **real-time transactional operations**, including order placement, stock verification, and dynamic menu retrieval.

Other Projects

[Blogs](#)

- DayStacks** — Smart productivity app with **agentic chatbot** task management, in-app YouTube embeddings. **50+ active users**.
- TripNow** — Full-stack **ride-booking** application inspired by real world platforms with more than **21 HTTP API endpoints**.
- CIDC** — Official IT club site for AIT Pune with **40+ regular interactions**.

TECHNICAL SKILLS

- Language:** JavaScript, C++
- Frontend:** HTML, CSS, React.js, Next.js
- Backend:** Node.js, Express.js, FastAPI, REST APIs
- Databases:** MongoDB, MySQL, Supabase
- AI & Automation:** Prompt Engineering, MCP, AI Agents, Workflow Automations
- Tools & Platforms:** Git, GitHub, Docker, Postman, Figma, Canva, VS Code, Linux (Ubuntu)
- CS Fundamentals:** Fundamentals of DSA, DBMS

KEY HIGHLIGHTS

- Finalist in **5 national hackathons:** Global Digital Health Summit, NASA Space Apps Challenge, Meta PyTorch OpenEnv, Virtual Labs, and Mumbai Hacks.
- Won **AlgoFusion-PartEX.AI Hackathon** conducted by PartXAI Medicare health company.
- Frontend Developer Intern at **Sniffi & Nexabots**, led UI development from design to production, building fully responsive interfaces.